PRACTICAL-04

TIME SERIES

Question-01

Input:

```
INPUT
data("BJsales")
print(BJsales)
plot(BJsales,main="BJsales Time Series",ylab="Sales",xlab="Month",col="orange")
time<-1:length(BJsales)
trend_fit<-lm(BJsales~poly(time,2))
plot(BJsales,main="BJsales with Polynomial Trend",ylab="Sales",xlab="Month",col="orange")
lines(time,predict(trend_fit),col="green",lwd=2)
residuals<-BJsales-predict(trend_fit)
#D
diff_BJsales<-diff(BJsales)
plot(diff_BJsales,main="Differenced BJsales",ylab="Differenced Sales",xlab="Month",col="blue")
library(tseries)
adf.test(diff_BJsales)
#E
par(mfrow=c(1,2))
acf(diff_BJsales,main="ACF of Differenced BJsales")
pacf(diff_BJsales,main="PACF of Differenced BJsales")
#F
library(forecast)
fit<-auto.arima(BJsales)
summary(fit)
#G
residuals<-residuals(fit)
par(mfrow=c(1,2))
plot(residuals,main="Residuals of Fitted Model",ylab="Residuals",xlab="Month",col="blue")
acf(residuals,main="ACF of Residuals")
#H
forecast_values<-forecast(fit,h=10)
plot(forecast_values,main="10-step Forcast for BJsales",ylab="Sales",xlab="Month")
#i
plot(BJsales,main="Actual vs Predicted BJsales",ylab="Sales",xlab="Month",col="blue")
lines(forecast_values$mean,col="red",lwd=2)
```

Output:

The dataset given here is BJsales

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OUTPUT

#1

print(BJsales)

Time Series:

Start = 1

End = 150

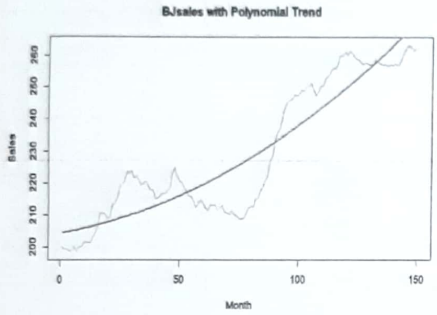
Frequency = 1

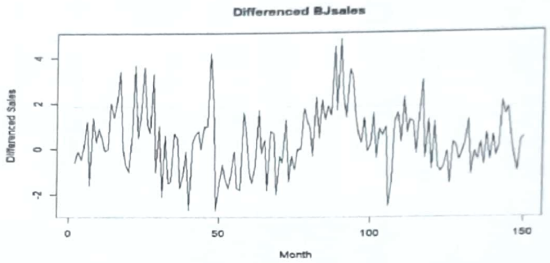
[1] 200.1 199.5 199.4 198.9 199.0 200.2 198.6 200.0 200.3 201.2 201.6 201.5 201.5 203.5 204.9 207.1 210.5

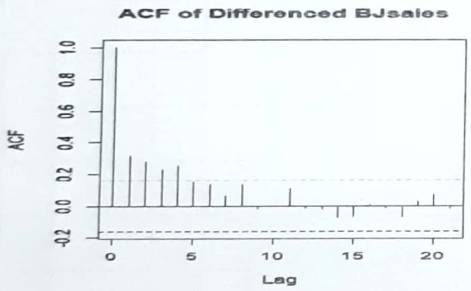
.....

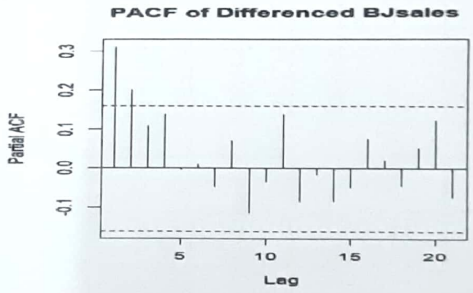
[120] 260.4 261.6 260.8 259.8 259.0 258.9 257.4 257.7 257.9 257.4 257.3 257.6 258.9 257.8 257.7 257.2 257.5

[137] 256.8 257.5 257.0 257.6 257.3 257.5 259.6 261.1 262.9 263.3 262.8 261.8 262.2 262.7









>summary(fit)

Series: BJsales

The first plot shows the original BJsales dataset with polynomial trend line.

The differenced BJsales ~~dat~~ plot shows the differenced BJsales data inorder to remove trends.

The ACF of differenced BJsales shows the autocorrelation function.

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PACF of Differenced plot shows the partial autocorrelation function plot of the differenced data.

ARIMA(1,1,1)

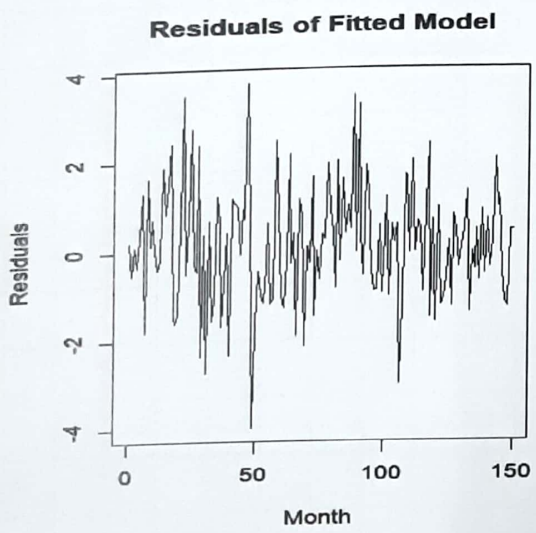
Coefficients:

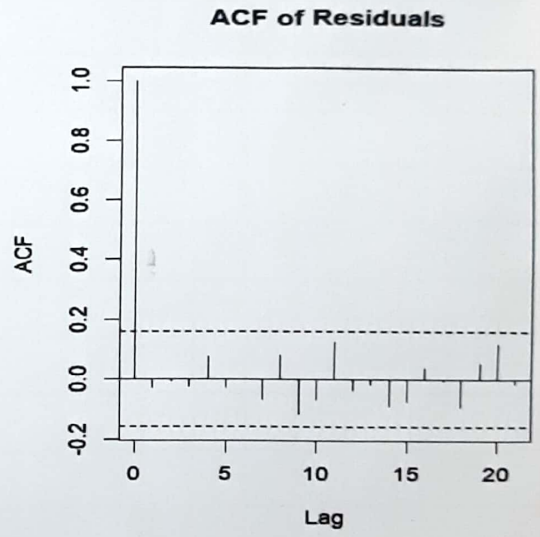
	ar1	ma1
	0.8800	-0.6415
s.e.	0.0644	0.1035

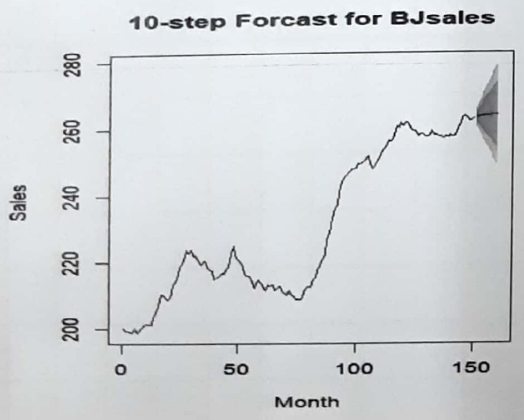
sigma^2 = 1.8: log likelihood = -254.37
AIC=514.74 AICc=514.9 BIC=523.75

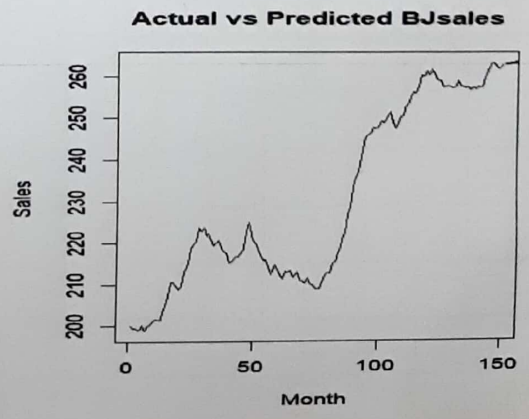
Training set error measures:

	ME	RMSE	MAE	MPE	MAPE	MASE	ACF1
Training set	0.1457572	1.328119	1.0447	0.06512899	0.4600935	0.8997702	-0.02622396









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f) ARIMA model is fitted for BJ sales where ar1 and ma1 explains the relationship between previous steps and error in predictions.

Mean error = 0.1457572
RMSE = 1.328119

The plot residuals of fitted models shows the difference between observed and fitted values over time.

ACF of residuals helps in diagnosing if the model fits well or not by showing the autocorrelation of residuals.

h) The 10-step forecast plot predicts the next 10 steps with confidence intervals. Here it shows that the sales have fluctuated but generally exhibit an upward trend.

(i) This graph shows the actual and predicted sales data in one graph.

Question 2:

Input:

```
#2
data("co2")
print(co2)
#b
plot(co2,main="co2 Time Series",ylab="Month",xlab="Year",col="purple")
#c
diff_trend<-diff(co2,differences = 1)
diff_seasonal<-diff(diff_trend,lag=12)
plot(diff_seasonal,main="Differenced Series(Stationary",ylab="Differenced
CO2",xlab="Time",col="green",type="o")
library(tseries)
adf.test(diff_seasonal)
#d
par(mfrow=c(1,2))
acf(diff_seasonal,main="ACF of Stationary Series")
pacf(diff_seasonal,main="PACF of Stationary Series")
```


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```
#e
library(forecast)
fit<-auto.arima(co2)
summary(fit)
#f
residuals<-residuals(fit)
par(mfrow=c(1,2))
plot(residuals,main="Residuals of Fitted Model",ylab="Month",xlab="Year",col="blue")
acf(residuals,main="ACF of Residuals")
#g
forecast_values<-forecast(fit,h=12)
plot(forecast_values,main="12-step Forcast for co2",ylab="Month",xlab="Year")
#h
plot(co2,main="Actual vs Predicted co2",ylab="Month",xlab="Year",col="purple")
lines(forecast_values$mean,col="red",lwd=2)
```

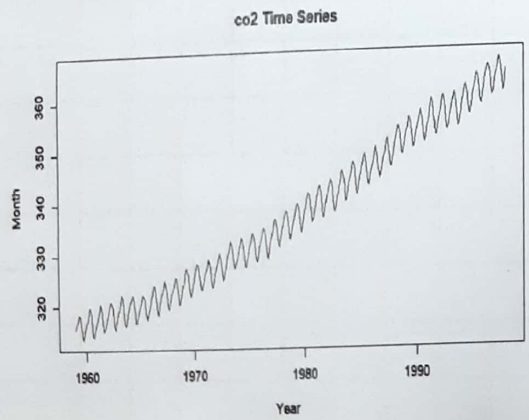
Output:

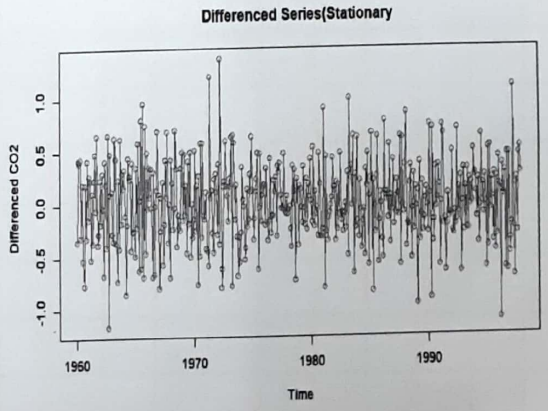
The ~~above~~ dataset "CO2" is about international atmospheric CO2 concentration data.

```
#2
print(co2)
     Jan   Feb   Mar   Apr   May   Jun   Jul   Aug   Sep   Oct   Nov   Dec
1959 315.42 316.31 316.50 317.56 318.13 318.00 316.39 314.65 313.68 313.18 314.66 315.43
1960 316.27 316.81 317.42 318.87 319.87 319.43 318.01 315.74 314.00 313.68 314.84 316.03
```

.....362.38 1997 363.23 364.06 364.61 366.40 366.84

365.68 364.52 362.57 360.24 360.83 362.49 364.34





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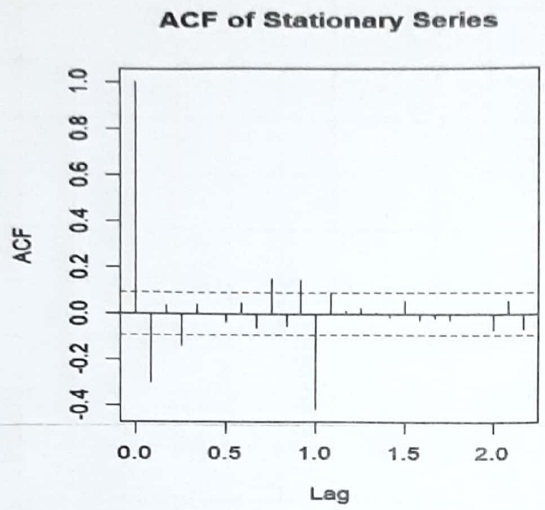
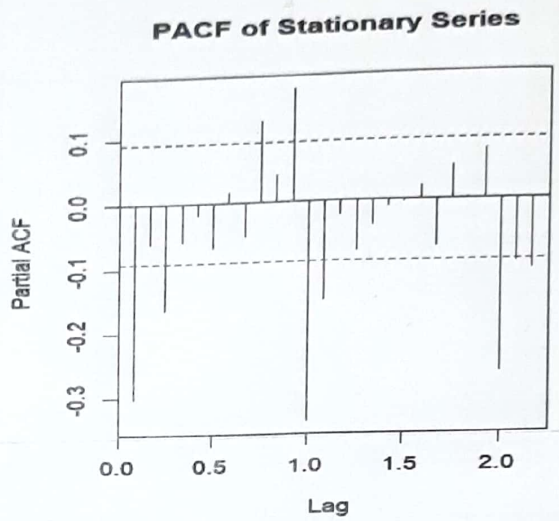
```
adf.test(diff_seasonal)

        Augmented Dickey-Fuller Test

data:  diff_seasonal
Dickey-Fuller = -8.9846, Lag order = 7, p-value = 0.01
alternative hypothesis: stationary
```



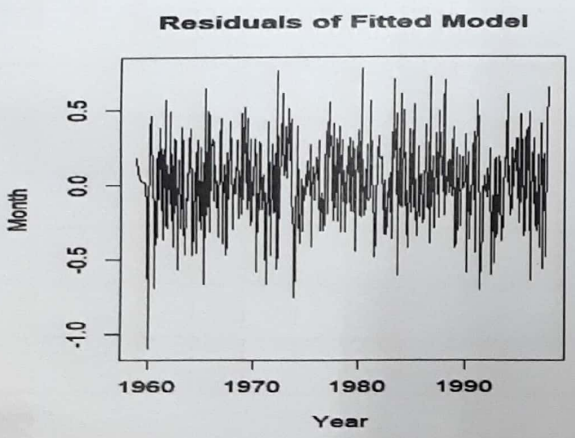


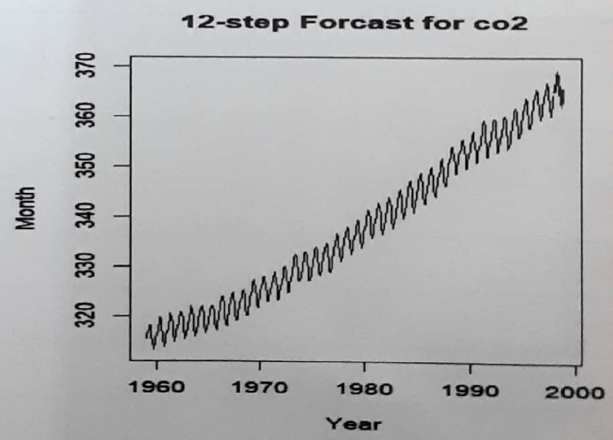

```
summary(fit)
Series: co2
ARIMA(1,1,1)(1,1,2)[12]

Coefficients:
         ar1     ma1    sar1    sma1    sma2
      0.2569 -0.5847 -0.5489 -0.2620 -0.5123
s.e.  0.1406  0.1204  0.5879  0.5701  0.4819

sigma^2 = 0.08576:  log likelihood = -84.39
AIC=180.78   AICc=180.97   BIC=205.5

Training set error measures:
                       ME    RMSE     MAE      MPE     MAPE     MASE      ACF1
Training set 0.01742092 0.287159 0.2303994 0.00507377 0.06845665 0.1819636 -0.00285809
```





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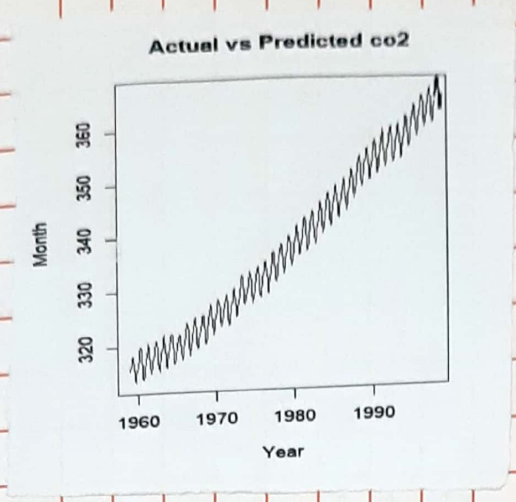

5) The ADF test checks for stationarity in the differenced series
Here the p-value 0.01 is less than 0.05 which implies that we reject H_0 i.e; the differenced series is stationary.

6) Here ARIMA model (1,1,1)(1,1,2)[12] was fitted. It shows how well the model fits the training data we have lower RMSE and MAE value which suggests a better fit and ACF1 value is close to zero it indicates that residuals are not significantly autocorrelated. This implies that the differenced series is stationary.

7) The CO2 time series graph indicates the long-term increase in atmospheric CO2.

The differenced series plot shows the differenced values of CO2 levels over time.

8) The ACF plot shows the autocorrelation function the lines represent the autocorrelation values at different lags.

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e) The PACF plot shows the correlation between the time series and its lagged values after removing the effects of shorter lags.

(f) Residuals of fitted model plot shows the residuals over the years. These residuals appears to fluctuate around zero without any pattern.

The ACF of residuals plot tells us that the residuals do not exhibit any significant correlation.

(g) The 12 step forecast plot helps to forecast CO_2 levels for next 12 steps (months) which represents the uncertainity in the CO_2 levels.

(h) This graph shows the actual and predicted CO_2 data.

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